LAPORAN PRAKTIKUM

Internet of Think



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TEKNOLOGI REKAYASA INTERNET

JURUSAN TEKNOLOGI INFORMASI

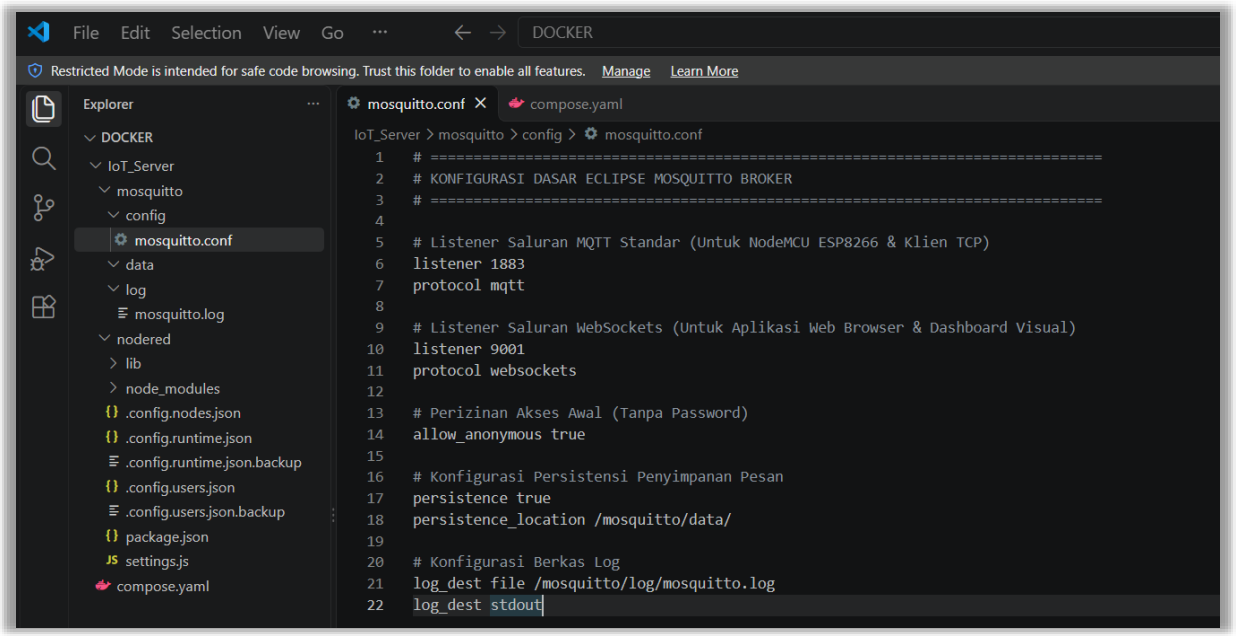
POLITEKNIK NEGERI LAMPUNG

2026/2027

Praktikum 1: Pembangunan Ekosistem IoT Server dengan Docker Compose

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2404:c0:5a10:dc6f:1f9c:54ac:8e22:6b63
   Temporary IPv6 Address. . . . . . : 2404:c0:5a10:dc6f:557d:afc0:68e8:cc2
   Link-local IPv6 Address . . . . . : fe80::de7c:50b7:5558:f511%17
   IPv4 Address. . . . . . . . . . . : 10.89.221.211
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::9800:d3ff:fedc:5cae%17
                                       10.89.221.173
```



mosquitto.conf | compose.yaml

IoT_Server > compose.yaml

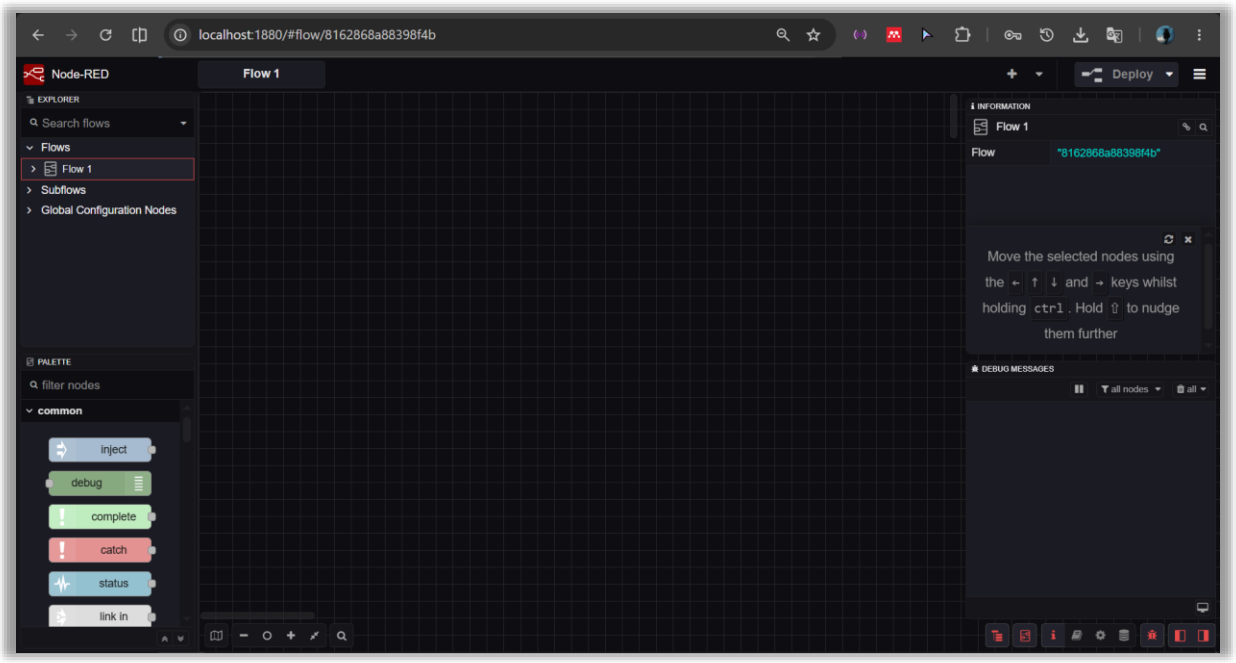
```yaml
services:
  # ============================================================
  # 1. Layanan Broker MQTT: Eclipse Mosquitto
  # ============================================================
  mqtt:
    image: eclipse-mosquitto:latest
    container_name: mqtt-broker
    restart: unless-stopped
    ports:
      - "1883:1883"  # Port MQTT murni untuk mikrokontroler
      - "9001:9001"  # Port WebSockets untuk dashboard web
    volumes:
      - ./mosquitto/config:/mosquitto/config
      - ./mosquitto/data:/mosquitto/data
      - ./mosquitto/log:/mosquitto/log

  # ============================================================
  # 2. Layanan Visual Rule Engine: Node-RED
  # ============================================================
  nodered:
    image: nodered/node-red:latest
    container_name: node-red
    restart: unless-stopped
    ports:
      - "1880:1880"  # Port antarmuka web builder Node-RED
    environment:
      - TZ=Asia/Jakarta
    volumes:
      - ./nodered:/data
    depends_on:
      - mqtt
```

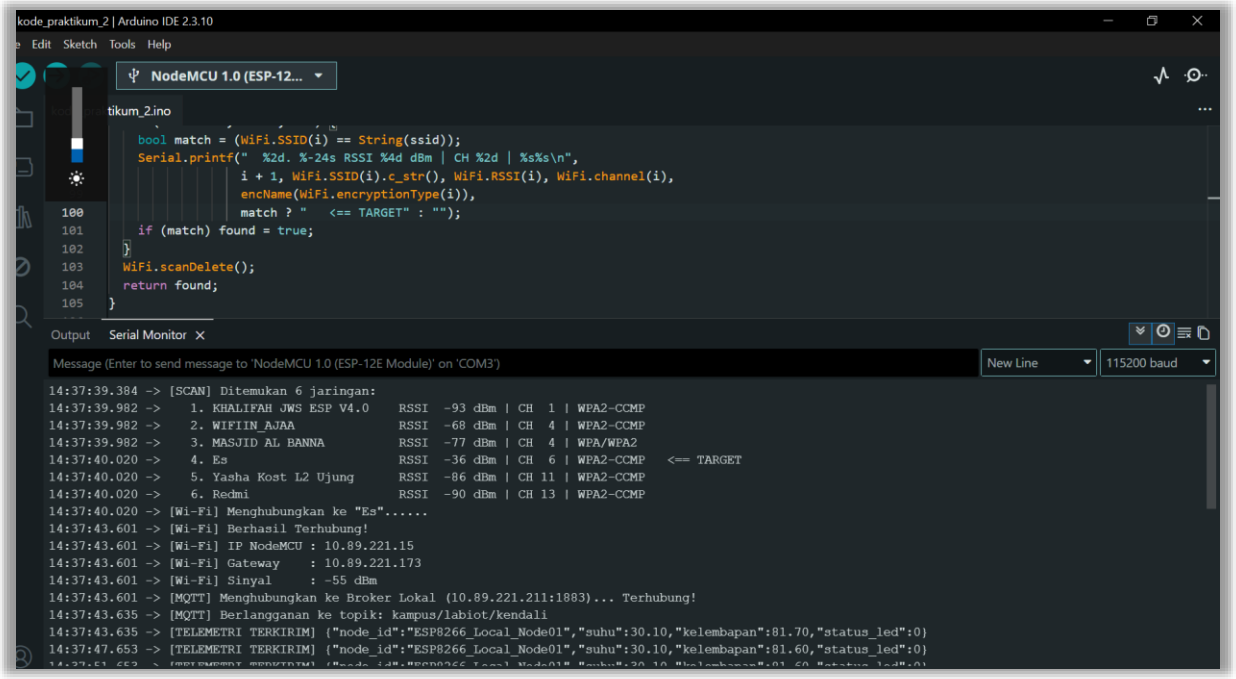
Menjalankan *docker compose up -d*

```
Windows PowerShell
PS D:\DOCKER\IoT_Server> docker compose up -d
[+] up 2/2
 ✔ Container mqtt-broker Running                                    0.0s
 ✔ Container node-red    Running                                    0.0s
PS D:\DOCKER\IoT_Server> docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED         STATUS
              PORTS
              NAMES
c35e9c1d0135   nodered/node-red:latest    "./entrypoint.sh"        3 minutes ago   Up 3 minut
es (healthy)   0.0.0.0:1880->1880/tcp, [::]:1880->1880/tcp
              node-red
3a2bb9f271b0   eclipse-mosquitto:latest   "/docker-entrypoint.…"   3 minutes ago   Up 3 minut
es             0.0.0.0:1883->1883/tcp, [::]:1883->1883/tcp, 0.0.0.0:9001->9001/tcp, [::]:9001
->9001/tcp   mqtt-broker
PS D:\DOCKER\IoT_Server>
```

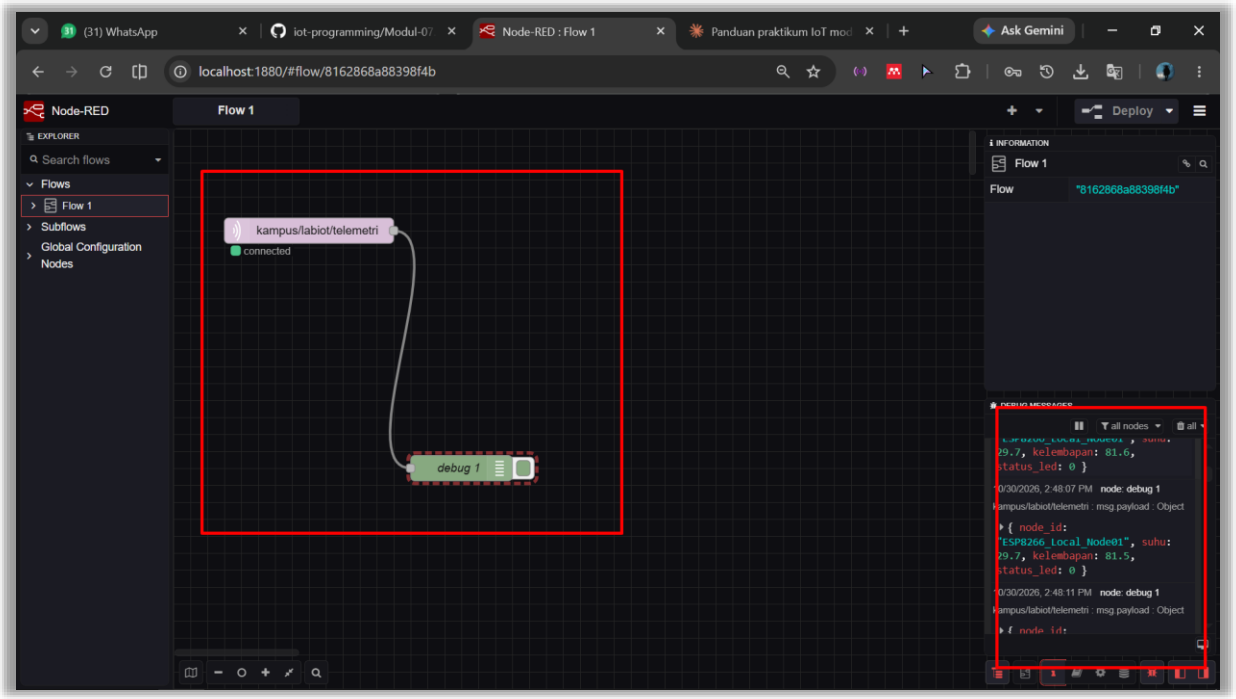
Setelah berhasil kemudian membuka link http://localhost:1880 adakah nodered



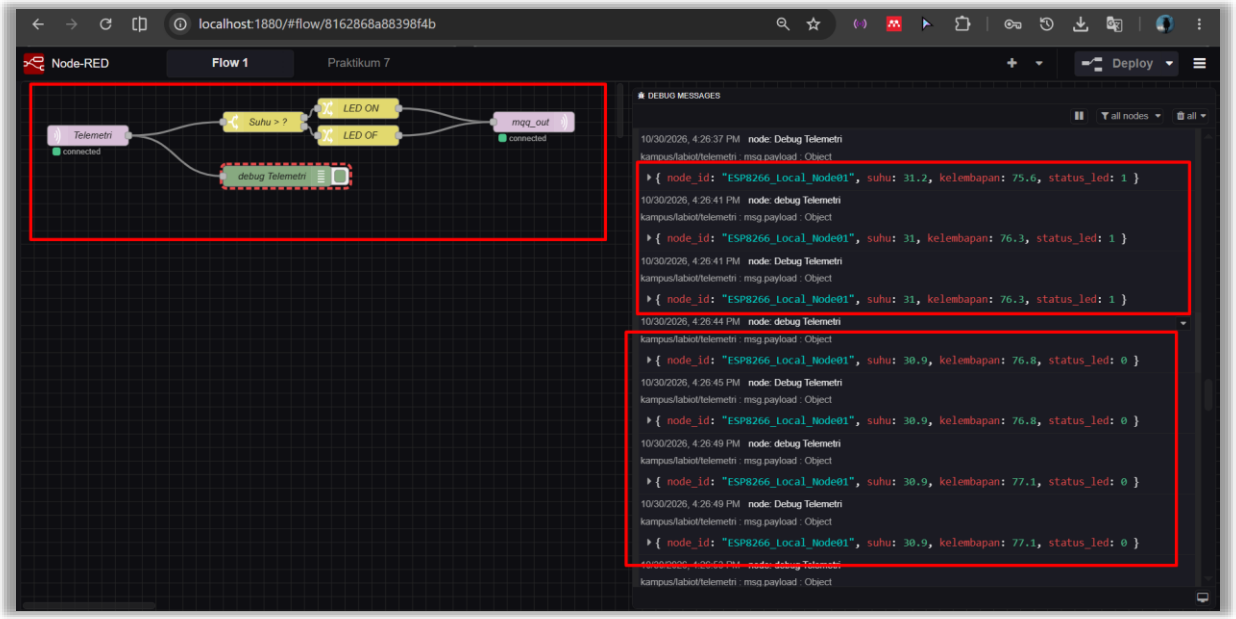
Praktikum 2: Migrasi Simpul NodeMCU ESP8266 ke Broker MQTT Lokal

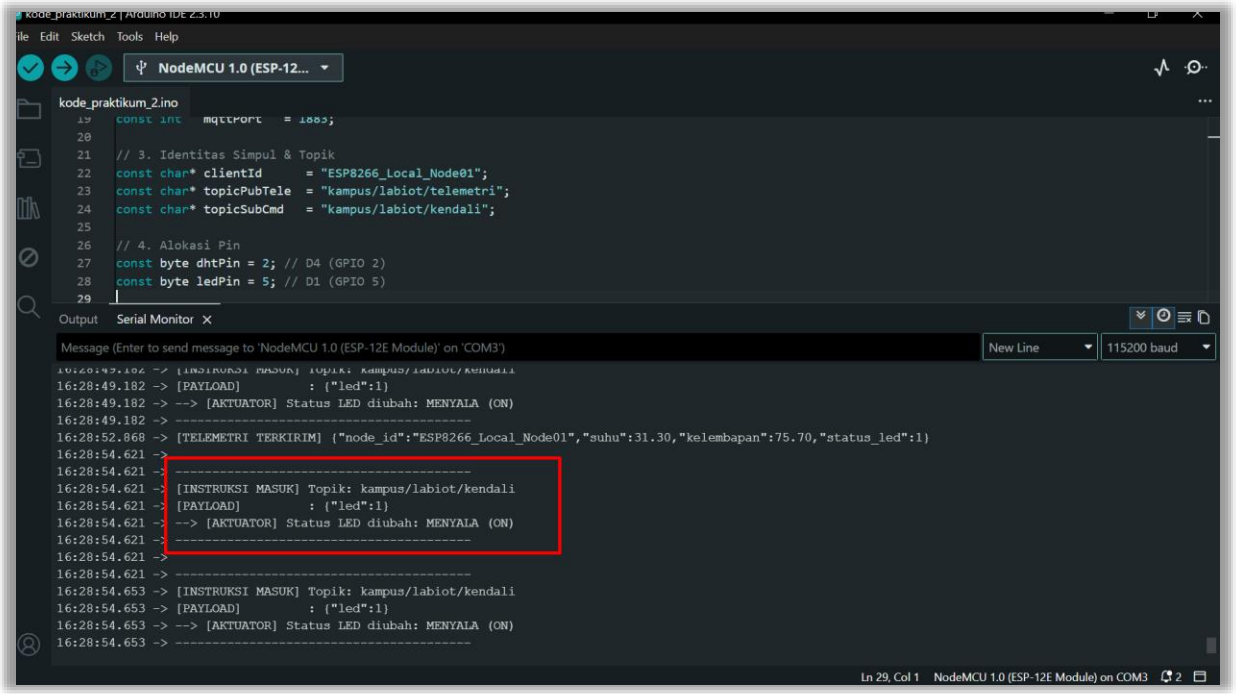


Praktikum 3: Integrasi Alur Data Telemetri Dua Arah pada Node-RED

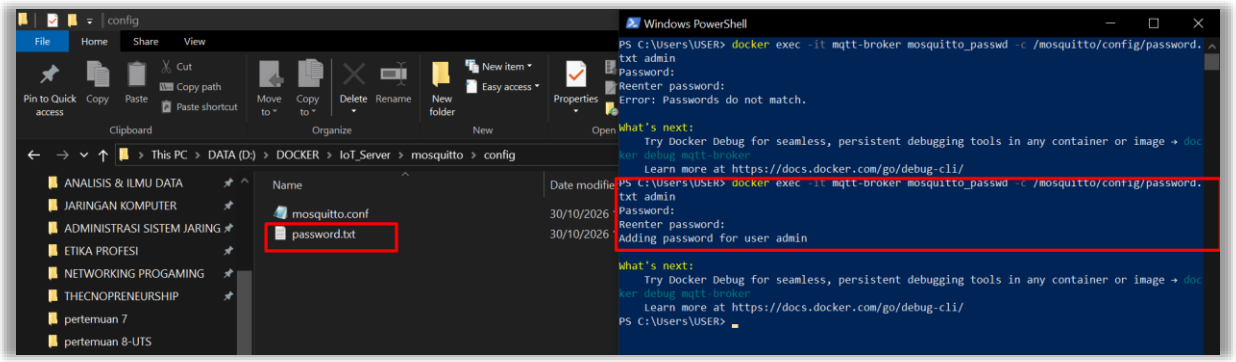


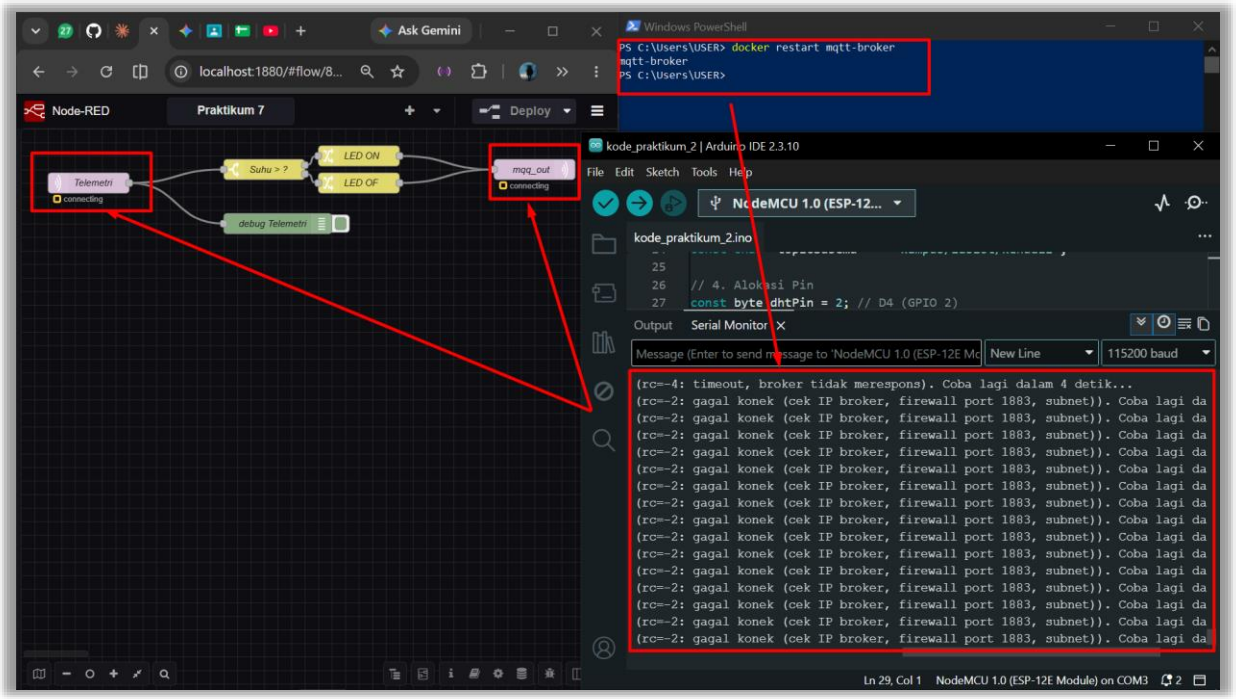
Praktikum 4: Implementasi Rule Engine Otomasi Aktuator Loop Tertutup



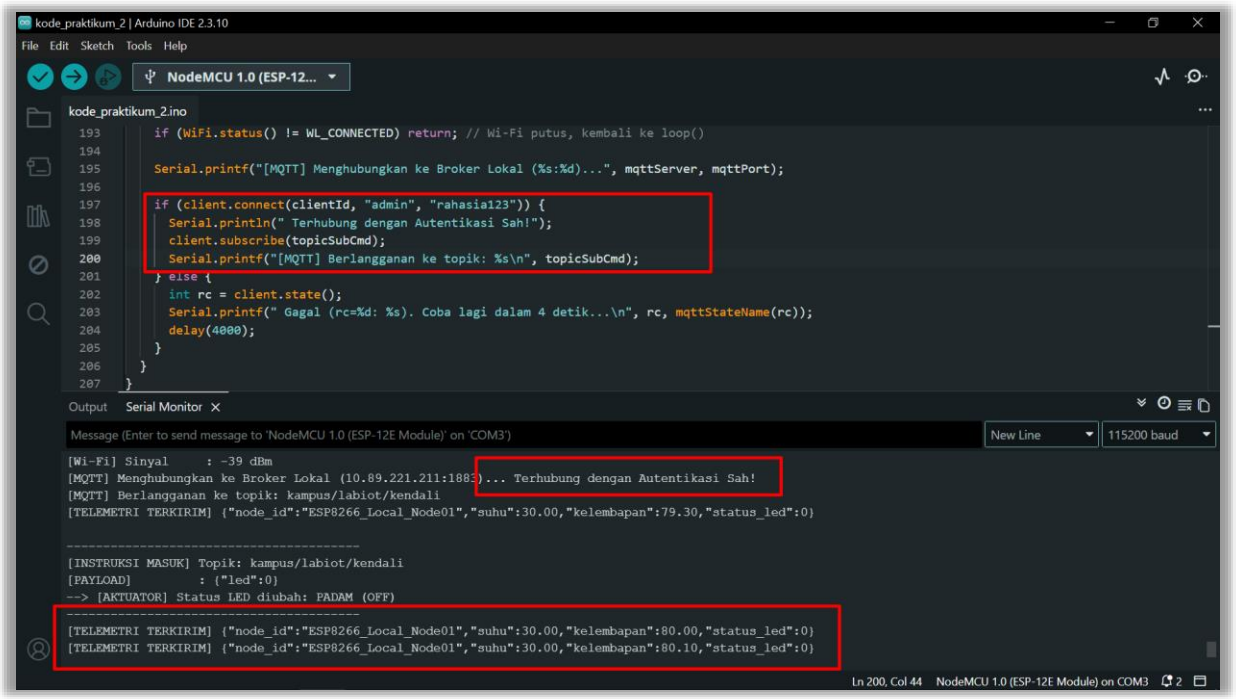


TUGAS : Pengamanan Server IoT Industri (MQTT Authentication & Access Control)

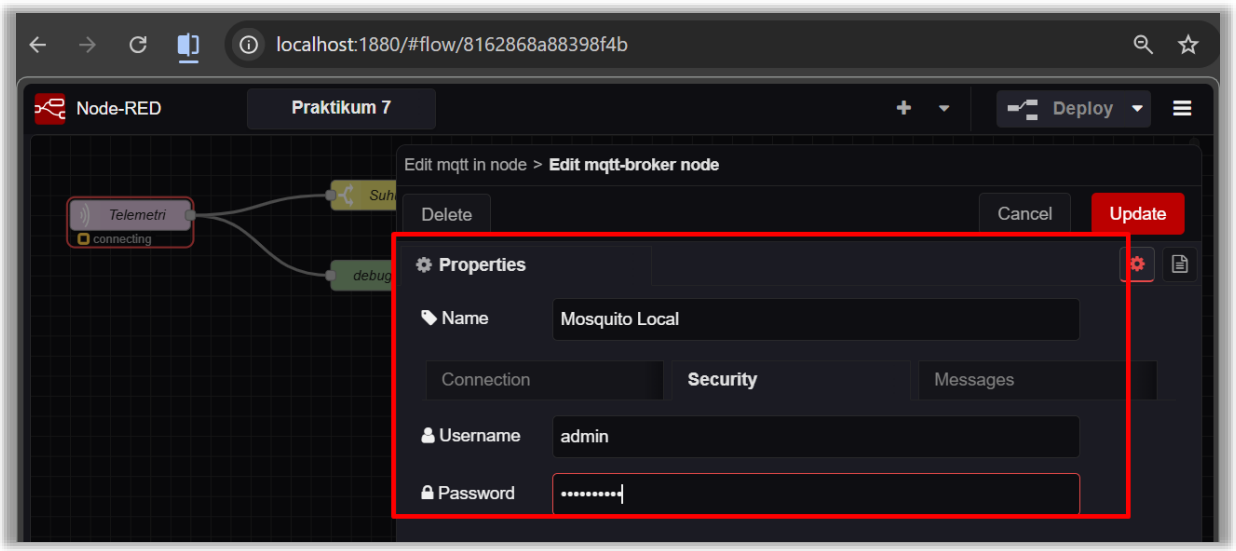


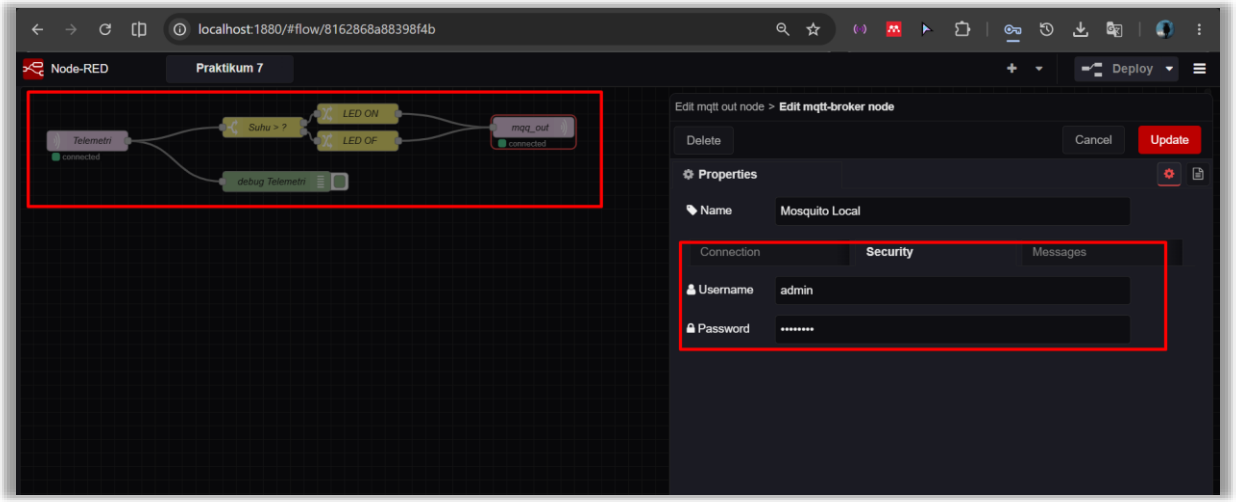


Memodifikasi kode agar bisa menggunakan username dan password



Login di nodered


localhost:1880/#flow/8162868a88398f4b
Node-RED
Praktikum 7
Deploy
Edit mqtt in node > Edit mqtt-broker node
Telemetri
connecting
debug
Delete
Cancel
Update
Properties
Name
Mosquito Local
Connection
Security
Messages
Username
admin
Password


localhost:1880/#flow/8162868a88398f4b
Node-RED
Praktikum 7
Deploy
Telemetri
connected
Suhu > ?
LED ON
LED OF
mqq_out
connected
debug Telemetri
Edit mqtt out node > Edit mqtt-broker node
Delete
Cancel
Update
Properties
Name
Mosquito Local
Connection
Security
Messages
Username
admin
Password